

Alex Evans

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Education

University of Oxford Oct. 2025 – Present

DPhil in Statistics and Materials

- **Research:** Establishing a forward and inverse model between the geometric and physicochemical properties of a capsomere set and the capsid architecture it assembles into, so that the display geometry of a vaccine scaffold becomes an input to design rather than an outcome of it. *Supervised by Prof. Charlotte Deane, Prof. Angus Kirkland and Prof. Maxim Ryadnov.*

Imperial College London Oct. 2023 – Oct. 2024

MRes Machine Learning in the Physical Sciences - Distinction

- **Research:** Applied diffusion and flow-based techniques in modelling quantum apparatus used in dark matter and gravitational wave research. *Supervised by Mikael Mieskolainen and Charles Baynham.*

University of Birmingham Oct. 2016 – June 2020

BEng Mechanical Engineering - Upper 2:1

Research

A Geometric and Physicochemical Model for Vaccine Scaffold Design Oct. 2025 – Present

University of Oxford

- Parameterising the design space of spherical capsid architectures as a Delaney–Dress symbol together with a metric realisation, one latitude, azimuth and radius per capsomere orbit, extending the categorisation beyond Caspar–Klug and quasi-equivalence to any protein nanocage. Collaborating with Prof. Reidun Twarock.
- Built a round-trip pipeline that reduces a deposited assembly to its capsomere interaction network, recovers the rotation group combinatorially before any geometry is fitted, then reads off the symbol and its realisation. Across 1,000 assemblies drawn from a 2,147-structure sweep of the PDB, every architecture rebuilds from its recorded parameters alone to within 1.1 Å of the measured capsomere centres, a quarter of them exactly.
- Working towards a normalised distribution over the architectures a given capsomere set can access, and its Bayesian inversion, returning the parameters and tolerances a complementary building block must satisfy to realise a chosen architecture, so polymorphism is scored before synthesis rather than observed after it.
- Techniques: Group Theory, Simulation-Based Inference, Cryo-EM Single Particle Analysis

AI Driven European Space Agency Satellite Retrieval June 2024 – Oct. 2025

National Physical Laboratory (NPL)

- Developed a generative AI model for the European Space Agency TRUTHS (Traceable Radiometry Underpinning Terrestrial and Helio-Studies) mission, aiming to calibrate the satellite’s onboard HIS (Hyperspectral Imaging Spectrometer) to retrieve solar irradiance from thermal readings to observe the effects of climate change mitigation policy.
- Modelled the encoded posterior through the use of a continuous-time diffusion model with transformer architecture for conditioning, enabling samples of the measurand to be generated and therefore providing reliable uncertainty quantification on predictions.
- Techniques: Diffusion; Key Libraries: JAX

Generative AI for Cold Atom Interferometry Nov. 2023 – Oct. 2024

Imperial College London

- Modelled quantum apparatus used in AION (Atom Interferometer Observatory and Network), a national project across various UK universities aiming to detect ultra-light dark matter and gravitational waves through cold-atom interferometry. This model facilitated the application of Bayesian inference to find the ideal set of parameters for optimal cooling.
- Ran 2^{19} atomic simulations with parameters sampled from a Sobol sequence on the Imperial College Lon-

don HPC servers, totalling ≈ 22 CPU-years of total computational time across 500 concurrent simulation instances.

- Applied normalising flows and diffusion in developing a model to generate cooled atoms for given sets of parameters, with generated data from the diffusion model being indistinguishable from the ground-truth to a random forest classifier.
- Developed novel techniques for both model classes: denoising the data over an initial set of epochs was found to significantly reduce manifold overfitting in normalising flows, and stabilising an approximation of the trace of the Hessian (Jacobian of the score) to mitigate it in diffusion models.
- Applied Bayesian optimisation on the simulation to find the optimal set of parameters to maximise an objective function derived from probability of transmission, average velocity and atom dispersion.
- Techniques: Diffusion, Normalising Flows, Deep Neural Networks, Decision Trees, Bayesian Optimisation; Key Libraries: JAX, PyTorch, HyperOpt, XGBoost

AI for Autonomous Marine Vessels

Jan. 2024 – June 2024

National Physical Laboratory (NPL)

- Collaborated with the UK Hydrographic Office on a project to develop a standardised language to be interpreted by autonomous marine vessels, making significant strides towards a machine-interpretable model of the S100 series of navigational charts.
- Developed an LLM workflow to translate tags from the IHO S-100 series of navigational charts into machine-readable format, with XML validation against an XSD schema.
- Techniques: Large Language Models; Key Libraries: Ollama

Employment

Scientist

Teddington, London

National Physical Laboratory (NPL)

Jan. 2024 – Oct. 2025

- Collaborated across Earth Observation & Climate (ESA TRUTHS) and Informatics (HADO/MAAT): generative AI for satellite radiometric calibration and assurance of autonomous systems (Heathrow security drones; maritime vessels).

Data Analyst

Teddington, London

National Physical Laboratory (NPL)

Nov. 2021 – Jan. 2024

- Utilised Azure cloud platform pipelines to create a unified SQL database of data from across the organisation, enabling finance and project management data to interact and therefore significantly improving the capability and quality of organisational reporting.

Awards

STEM for Britain Finalist

March 2025

Department for Science, Innovation and Technology

- Selected as a finalist for STEM for Britain 2025, presenting research on Generative AI for Cold Atom Interferometry at parliament to MPs and judges from academia.

Academic Achievement

Oct. 2024

Imperial College London

- Achieved top 3 final grade in cohort; ranked 1st in Statistics for Experimental Physics and Self-Study Project on score-matching and diffusion in medical imaging.

Courses

Probabilistic Machine Learning Summer School

July 2025

University of Cambridge

- Selected for and attended a summer school at Cambridge University, covering various Bayesian methods and deep generative models.